

Michigan Tech

Background Math

Matrices & ODEs & parameters, Oh My!

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Math Sciences

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Chaotic Jerk IVPS from Sprott

No.	Jerk ODE	IC $(y, \dot{y}, \ddot{y})$	Lyapunov exps
1	$\ddot{y} = -2.017\ddot{y} \pm \dot{y}^2 - y$	$(0, 0, \pm 1)$	$0.055, 0, -2.072$
2 DM	$\ddot{y} = -2.8\dot{y} \pm y + y^2$	$(\mp 0.5, -1, 1)$	$0.002, 0, -0.002$
3	$\ddot{y} = -0.44\ddot{y} - 2\dot{y} \pm (y^2 - 1)$	$(0, 0, 0)$	$0.105, 0, -0.545$
4	$\ddot{y} = -0.5\ddot{y} - \dot{y} \pm y \pm y^2$	$(0, \pm 1, 0)$	$0.094, 0, -0.594$
5	$\ddot{y} = -2\dot{y} \pm (y - 1)$	$\pm(-1, -1, 1)$	$0.003, 0, -0.003$
6	$\ddot{y} = -0.6\ddot{y} - \dot{y} \pm (y - 1)$	$(0, 0, 0)$	$0.036, 0, -0.636$
7 OS	$\ddot{y} = -0.3\ddot{y} - 0.3\dot{y} - D(y) + 1$	$(0, 0, 0)$	$0.042, 0, -0.342$
8	$\ddot{y} = -0.3\ddot{y} - 0.3\dot{y} - R(y) - 1$	$(0, 0, 0)$	$0.042, 0, -0.342$
9	$\ddot{y} = -2.9\dot{y} \pm (0.7y - D(y) + 1)$	$\pm(0, -0.5, 0.5)$	$0.005, 0, -0.005$
10	$\ddot{y} = -2.9\dot{y} \pm (0.7y - R(y) - 1)$	$\pm(0, -0.5, 0.5)$	$0.005, 0, -0.005$
11 LB	$\ddot{y} = -0.5\ddot{y} - \dot{y} - y + \operatorname{sgn}(y)$	$(0, 0, 0)$	$0.080, 0, -0.580$
12	$\ddot{y} = -0.5\ddot{y} - \dot{y} + y - \operatorname{sgn}(y)$	$(0, 0, 0)$	$0.080, 0, -0.580$

Jerk IVPS from Sprott

No.	Jerk ODE	IC $(y, \dot{y}, \ddot{y})$	Lyapunov exps
13	$\ddot{y} = -0.7\ddot{y} - \dot{y} - y + H(y)$	$(0, 0, 0)$	0.035, 0, -0.735
14	$\ddot{y} = -0.4\ddot{y} - \dot{y} - y + 2S(y)$	$(0, 0, 0)$	0.050, 0, -0.450
15 AP	$\ddot{y} = -0.4\ddot{y} - \dot{y} + y - 2S(y)$	$(0, 0, 0)$	0.050, 0, -0.450
16 IZ	$\ddot{y} = -0.19\ddot{y} - \dot{y} - y + 2 \tanh y$	$(0, 0, 0)$	0.100, 0, -0.290
17	$\ddot{y} = -0.19\ddot{y} - \dot{y} + y - 2 \tanh y$	$(0, 0, 0)$	0.100, 0, -0.290
18 AW	$\ddot{y} = -3.7\dot{y} \pm (y - y^3)$	$(0, 0, 0)$	0.020, 0, -0.020
19 BF	$\ddot{y} = -0.6\ddot{y} + 2.8\dot{y} - \dot{y}^3 - y$	$(0, 0, 0)$	0.110, 0, -0.710
20 AR	$\ddot{y} = -0.7\ddot{y} - \dot{y} + y - y^3$	$(0, 0, 0)$	0.100, 0, -0.800
21 EB	$\ddot{y} = -0.35\ddot{y} - \dot{y} - y + y^3$	$(0, 0, 0)$	0.100, 0, -0.450
22 TA	$\ddot{y} = -0.2\ddot{y} - \dot{y} \pm \sin y$	$(0, 0, 0)$	0.120, 0, -0.320

Jerk ODEs from Sprott

$$\text{Diode } D(y) = \begin{cases} y & y \geq 0, \\ 0 & y < 0, \end{cases} = \max(y, 0)$$

$$\text{Relay } R(y) = \begin{cases} 1 & y \geq 0, \\ -1 & y < 0, \end{cases} = \text{sgn}(y)$$

$$\text{Saturation } S(y) = \begin{cases} -1 & y < -1, \\ y & |y| \leq 1, \\ 1 & y > 1, \end{cases} = \max\{-1, \min(y, 1)\}$$

$$\text{Heaviside } H(y) = \begin{cases} 1 & y > 0, \\ 0 & y \leq 0 \end{cases} = (\text{sgn}(y) + 1) / 2$$

Numerical Solve and Plot

Standard built-in ODE solvers all solve systems of first order ODEs with appropriate initial conditions for a vector valued solution $y(t)$

$$y' = f(t, y) \quad y(t_0) = x.$$

Solution are usually represented by vectors $y_1, y_2, \dots, y_N$ at times $t_1, t_2, \dots$

- Mathematica can directly solve higher order matrix ODEs and returns interpolating functions. It's heuristics usually select suitable algorithms.
- Julia does not solve higher order systems directly or automatically select algorithms. Julia does solve matrix valued first order ODEs.
- I believe every other solver returns lists of data for first¹ order systems.

¹Special 2nd order codes exist.

Summary

- It is fairly easy to numerically solve IVP.
- Most solvers need the function $f(t, y)$ for the ODE

$$y' = f(t, y)$$

- Most solvers return a list of vectors approximating $y_1, y_2, \dots, y_N$ the solution $y(t)$ at a sequence of points $t_1, t_2, \dots, t_N$.
- It is not hard to rewrite a higher order ODE as a system.